

SIMATS ENGINEERING
ASSESSMENT TOOL 1: Analytical Problem Solving

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA15

COURSE OUTCOME COVERED

CO2: *Analyze virtualization architectures to identify performance and security issues in cloud computing environments. (BL3)*

Dave's Taxonomy: Manipulation (Level 2)
Question Count: 5

Total Marks: 50

Code of Conduct

I M. AJAY / 192421021 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: AP 07

Assessment Tasks

Analytical Problem Solving

1. Compare Type-1 and Type-2 hypervisors for a cloud datacenter hosting mission-critical applications. Analyze performance, security, and manageability, then justify the better choice.
2. A cloud provider must isolate workloads belonging to finance, healthcare, and student portals on the same hardware. Analyze how virtualization architecture supports isolation and identify two major attack surfaces.
3. Study the following VM host utilization snapshot and recommend corrective actions: CPU 92%, memory 88% storage I/O latency 35 ms, average VM boot time 170 s. Explain the likely bottlenecks.
4. Analyze how Software Defined Datacenter architecture improves agility compared with a traditional datacenter. Support your answer with compute, storage, and network virtualization considerations.
5. A multi-cloud federation project fails due to identity mismatches between providers. Analyze the issue and propose a secure identity and privacy design using federation concepts.

Type-1 Hypervisor

Runs directly on hardware

High performance

More secure

used in data centers

Type-2 Hypervisor

Runs on a host operating system

Lower performance

Less secure

used for testing and personal use.

Justification:

For mission-critical cloud applications, Type-1 Hypervisor is the better choice because it offers better performance, stronger security, and easier management. Ex: Includes Xen, Microsoft Hyper-V, and KVM.

Virtualization creates separate virtual machines (VMs) on the same hardware. Each VM has its own OS, CPU, and storage, preventing one workload from affecting another.

Two major attack surfaces:

* Hypervisor attacks - Attackers exploit vulnerabilities in the hypervisor to access other VMs.

- * Virtual network attacks - weak virtual network config may allow unauthorized communication between VMs.

Conclusion: proper VM isolation, firewall rules, and regular security updates improve protection.

Given:

- * CPU: 92%.
- * Memory: 88%.
- * Storage I/O latency: 35ms
- * VM Boot Time: 170s.

Likely Bottlenecks:

- * High CPU usage causes slow processing
- * High memory usage may lead to swapping
- * High storage latency delays disk operations.
- * Long boot time indicates storage or resource.

Corrective Actions:

- * Add more CPU and RAM.
- * Upgrade to SSD / NVMe storage
- * Balance workloads across hosts
- * Optimize VM resource allocation.

An SDDC virtualizes compute, storage, and networking, allowing centralized software management.

Advantages:

- * Compute Virtualization: Quickly creates and manages VMs.
- * Storage Virtualization: Allocates storage dynamically
- * Network Virtualization: Simplifies network configuration and resource utilization compared to traditional data centers.

Identity mismatches occur when different cloud providers use different user authentication systems.

Secure Identity and Privacy Design:

- * Use single sign-on (SSO) with federated identity
- * Implement standard protocols like SAML or OAuth 2.0 / OpenID Connect.
- * Enable Multi-Factor Authentication (MFA).
- * Apply Role-Based Access Control (RBAC) and encrypt sensitive data

Conclusion:

Federated identity management provides secure authentication, protects user privacy, and enables

smooth access across multiple cloud providers